17. $\lim_{t\to -1}\left(\frac{2t^5 - t^4}{5t^2+4}\right)^3$

$= \left[\lim_{t\to -1} \frac{-3}{9}\right]^3$

$= -\frac{1}{27}$

19. $\lim_{x\to -2} \frac{x^2-x-6}{3x^2+5x-2}$

$= \lim_{x\to -2} \frac{(x-3)(x+2)}{(3x-1)(x+2)}$

$= \lim_{x\to -2} \frac{x-3}{3x-1}$

$= \frac{5}{7}$

19. $\lim_{t\to 3} \frac{t^3-27}{}$

25. $$\lim_{x\to\infty} \frac{\sqrt{1+4x^6}}{2-x^3}$$

$$= \lim_{x\to\infty} \frac{\sqrt{x^6}\sqrt{\frac{1}{x^6}+4}}{x^3(\frac{1}{x^3}-1)}$$

$$= \lim_{x\to\infty} \frac{\sqrt{\frac{1}{x^6}+4}}{\frac{1}{x^3}-1}$$

$$= \frac{2}{-1} = -2$$

26. $$\lim_{x\to-\infty} \frac{\sqrt{1+4x^6}}{2-x^3}$$

$$= \lim_{x\to-\infty} \frac{\sqrt{x^6}\sqrt{\frac{1}{x^6}+4}}{x^3(\frac{2}{x^3}-1)}$$

$$= \lim_{x\to-\infty} \frac{|x^3|\sqrt{\frac{1}{x^6}+4}}{x^3(\frac{2}{x^3}-1)}$$

$$= -\frac{2}{-1} = 2$$

△

29. $$\lim_{t\to\infty}(\sqrt{25t^2+2}-5t)$$

$$= \lim_{t\to\infty}(\sqrt{25t^2+2}-5t) \times \frac{\sqrt{25t^2+2}+5t}{\sqrt{25t^2+2}+5t}$$

$$= \lim_{t\to\infty} \frac{25t^2+2-25t^2}{\sqrt{25t^2+2}+5t}$$

$$= \lim_{t\to\infty} \frac{2}{\sqrt{25t^2+2}+5t}$$ → 分母不用再化简 or 提取 x
∵ 提取 x 的目的是约分

$$= 0$$

30. $$\lim_{t\to-\infty}(\sqrt{4x^2+3x}+2x)$$

$$= \lim_{t\to-\infty}(\sqrt{4x^2+3x}+2x) \times \frac{\sqrt{4x^2+3x}-2x}{\sqrt{4x^2+3x}-2x}$$

$$= \lim_{t\to-\infty} \frac{4x^2+3x-4x^2}{\sqrt{4x^2+3x}-2x} = \lim_{t\to+\infty} \frac{3x}{|x|\sqrt{4+\frac{3}{x}}-2x} = \lim_{t\to-\infty} \frac{3}{-\sqrt{4+\frac{3}{x}}-2}$$

$$= -\frac{3}{4}$$

Use IVT

$\ln x = x - \sqrt{x}$ $(2,3)$

$\Delta =$

$\ln x$ is continuous at $x \in (0,\infty)$, x is continuous on $\mathbb{R}$, and $\sqrt{x}$ is continuous at $x \in [0,+\infty)$, so $f(x) = \ln x - x + \sqrt{x}$ is continuous at $x \in (2,3)$

$$\left.\begin{array}{l} f(2) = \ln 2 - 2 + \sqrt{2} = 0.107 \\ f(3) = \ln 3 - 3 + \sqrt{3} = -0.169 \end{array}\right]$$

$\because f(2) > 0$ and $f(3) < 0$ and $f(x)$ continuous at $x \in (2,3)$

$\therefore$ There is $f(x) = 0$

结论!!! There exists of at least one number $x = c$, where $c \in (2,3)$, such that $f(c) = 0$